

Theory Lab: Morphological Classification & The Physical Signature Hypothesis

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Protocol: CP-002-O-D-JNP v2.0 (Janus SME)

Status: HYPOTHESIS — Requires experimental validation

Context: DIMA directive to expand TEM from single-file scanner to holistic classification platform

The Four Capabilities DIMA Is Requesting

#	Capability	Current Status	Difficulty
1	Holistic directory scanning with coverage guarantee	v1.0 CLI has batch <dir> — needs hardening + reporting	Medium
2	Morphological classification — group files by shape, not extension	Not implemented — the "Morphological Encyclopedia" (IMG_2870) was envisioned but not built	Hard
3	Codeless malware identification — identify malware TYPE from geometry alone	Theoretical — depends on whether physical signatures exist	Research Required
4	Research corpus — acquire inert malware samples to test the hypothesis	Not started	Operational

These are **not independent features**. They form a pipeline:



↑
If signatures exist,
accentuate in mugshot

You must start at the bottom (corpus) to know if the top (classification) is even possible.

The Physical Signature Hypothesis

The Question

Do fully unmasked malware files have a morphological signature — visible in their TEM fingerprint — that categorically distinguishes them from ALL regular files, even encrypted ones?

Why This Might Be True (CORPUS Domain)

Every file type has structural constraints that its bytes must satisfy:

File Type	Structural Constraint	Expected Morphology
Plain text (.txt, .csv)	ASCII range (0x20-0x7E dominant), line-delimited	Low entropy, linear Markov chains, smooth torus
Compressed (.zip, .gz)	Near-uniform byte distribution, header + payload structure	High entropy body, low entropy header — bimodal
Encrypted (.aes)	Near-perfect uniform distribution, no structure at any scale	Maximum entropy, perfectly smooth donut
PE executable (.exe)	DOS header + PE header + sections (.text, .data, .rsrc) with distinct entropy profiles	Sectioned — each section has different entropy. Complex TCGE graph.
Ransomware	Encryption routines + key management + ransom note template + C2 comms	Encryption code (structured) + encrypted payload (smooth) = hybrid morphology
RAT	Network socket code + command parser + stealth routines + payload	Complex graph topology, high cross-reference density, backdoor intent flags
Dropper	Small bootstrap + large encrypted secondary payload	Extreme bimodal : tiny structured header, massive high-entropy body
Worm	Self-replication logic + propagation vectors + payload	Recursive graph structure (self-referencing), replication-signature topology

The Key Insight

Regular files are structurally *homogeneous* — a text file is text throughout, a ZIP is compressed throughout, an image follows its codec throughout.

Malware is structurally *heterogeneous* — it **MUST** contain multiple functional regions that serve different purposes (evasion, execution, payload delivery, communication). These regions have **different entropy profiles, different topological complexity, and different intent scores**.

This heterogeneity should be visible in the TEM morphology as:

- **Entropy discontinuities** (TFEA) — sudden jumps between high and low entropy regions
- **Graph clustering** (TCGE) — distinct functional modules with cross-references
- **Intent contradictions** (AISE) — benign-looking sections adjacent to aggressive ones
- **Zugzwang spikes** — the multi-domain disagreement that signals adversarial structure

[!IMPORTANT]

The hypothesis: Malware's morphological signature is not a single feature — it is the **pattern of discontinuities** across its structure. A clean file has a consistent morphology. Malware has a morphology that *contradicts itself internally*. This is the Zugzwang at the sub-file level.

Why Encrypted Files Are Different

DIMA specifically asked: can we distinguish malware from encrypted files?

- **Pure encrypted file** (.aes, .gpg): Near-uniform byte distribution **everywhere**. No structure at any scale. Smooth donut.
- **Malware with encrypted payload:** The payload section is high-entropy, BUT the bootstrap/loader code that unpacks it is **structured, low-entropy, and functionally complex**. The file has a "seam" between the structured loader and the encrypted payload.

That seam is the signature. A legitimately encrypted file has no seam — it's uniformly encrypted. Malware with an encrypted payload has a structural discontinuity where the loader meets the payload. The TEM should detect this as:

- A sharp entropy gradient in the TFEA sliding window
- A topological "bridge" in the TCGE graph connecting the loader subgraph to the payload region
- An AISE intent mismatch between the passive encrypted region and the active loader code

Visualization: Accentuating the Signature in the Mugshot

If the hypothesis holds, the mugshot accentuation would work like this:

Torus Fingerprint (Step 2A):

- Discontinuity regions rendered as **ridges or spikes** on the torus surface
- Color-coded: **--wire-primary** (cyan) for structure, **--heat-high** (red) for entropy spikes, **--heat-critical** (orange) for the SEAM between them

Superformula Mugshot (Step 2B):

- The Gielis superformula parameters already produce different shapes for different morphologies
- Accentuation: the $n_1/n_2/n_3$ parameters could be weighted to **exaggerate** the features that distinguish malware classes
- Ransomware → "hedgehog" (structured core with encrypted spines)
- RAT → "octopus" (central hub with tentacle-like network connections)
- Dropper → "egg" (smooth shell containing something hidden)
- Worm → "spiral" (self-similar recursive structure)

These are hypothetical morphologies — **they must be validated experimentally.**

Research Plan: Validating the Hypothesis

Phase R1: Corpus Acquisition

Goal: Collect at minimum 200 samples across 6 categories:

Category	Target Count	Source
Clean text files (.txt, .py, .md, .json)	50	Local filesystem, GitHub repos
Clean binaries (.exe, .dll — known-good)	30	Windows system files, signed binaries
Compressed/encrypted (.zip, .gz, .aes)	30	Generate locally using standard tools
Ransomware (defanged/inert)	30	MalwareBazaar, theZoo (GitHub), VirusTotal
RATs / Backdoors (defanged/inert)	30	MalwareBazaar, theZoo
Droppers / Loaders (defanged/inert)	30	MalwareBazaar, theZoo

Inert malware sources (legitimate research):

- **MalwareBazaar** (bazaar.abuse.ch) — free, curated, tagged by malware family
- **theZoo** (github.com/ytisf/theZoo) — live malware samples for research, password-protected ZIPs
- **VirusShare** (virusshare.com) — requires registration, massive corpus
- **Contagio** (contagiodump.blogspot.com) — curated by a security researcher

[!CAUTION]

Safety protocol: All malware samples must be handled in the Quarantine Zone ONLY. Never unzip on a production machine. The TEM scanner reads raw bytes — it does NOT execute files. But the samples themselves are dangerous outside the scanner.

Recommendation: Use the HP machine with a dedicated quarantine partition. Or better — a VM snapshot that can be rolled back.

Phase R2: Baseline Morphology Atlas

Goal: Run TEM on all 200 samples. Record the full TEMReport for each. Build a database:

For each sample:

- File hash (SHA-256)
- Declared file type (extension)
- Actual morphological class (TEM-derived)
- TFEA: global entropy, entropy profile, compression ratio
- TCGE: block count, edge density, cyclomatic complexity
- AISE: intent score, flags
- Markov: fingerprint hash, bigram entropy
- Zugzwang: Z-score (inter-module disagreement)
- CQSF: composite score, structural anomaly index

Phase R3: Cluster Analysis

Goal: Do the morphological fingerprints cluster by malware type?

Without ML (staying Tesla Principle — first principles):

1. Compute pairwise **cosine similarity** of the feature vectors
2. Project into 2D using the existing spectral embedding (same technique as TCGE graph layout)
3. Color by known category
4. **If clusters form** → the hypothesis holds. Different malware types have distinct physical signatures.
5. **If clusters don't form** → the hypothesis fails at this resolution. Either need finer-grained features or the signature doesn't exist.

Phase R4: Signature Extraction

If Phase R3 succeeds:

1. Identify which TEM features **most strongly** separate each cluster
2. Define a **Morphological Taxonomy** — named categories based on shape, not file type
3. Build the accentuation parameters: for each category, define which mugshot parameters to exaggerate
4. Implement in the v4.0 visualization as a "classification overlay"

Capability 1: Holistic Directory Scanning

Current State

The v1.0 CLI already has `python -m tem batch <directory>`. But "batch" is just "scan each file, print results." DIMA wants:

- **Coverage guarantee** — every file in the directory tree is scanned, with a manifest
- **Scan logging** — persistent record of what was scanned, when, and what the verdict was
- **Classification aggregation** — directory-level summary ("this directory contains 47 clean files, 3 suspicious, 1 quarantine-recommended")

Design Sketch

Directory Scan Report: C:\Users\Shadow\Downloads\suspicious_archive\

Coverage: 127/127 files scanned (100%)

Scan time: 14.3 seconds

Classification Summary:

Morphological Class	Count	File Types Observed
🟢 Text/Structured	89	.py, .json, .md, .txt
🟡 Compressed	21	.zip, .gz, .tar
🟠 Encrypted	8	.aes, .gpg, .enc
🟡 Hybrid-Bimodal	6	.exe, .dll
🔴 ZUGZWANG	3	.php, .exe, .dll

Note: Classification is by MORPHOLOGY, not extension.

"File Types Observed" shows what extensions happen to appear in each class.

⚠️ 3 files triggered Zugzwang detection:

1. update_helper.exe — Z=0.47 — AISE ↔ TFEA contradiction
2. config.php — Z=0.39 — TCGE ↔ TFEA contradiction
3. runtime.dll — Z=0.51 — AISE ↔ TCGE contradiction

The Key Inversion

Notice the table above: classification is by **morphology** (what the file looks like to TEM), and file extension is tracked as a secondary metric. This is DIMA's directive inverted:

| **Traditional:** "This is a .exe file. Is it malicious?"

| **TEM:** "This file has Hybrid-Bimodal morphology. It happens to have a .exe extension. Its Zugzwang score is elevated."

If a .txt file shows up in the "Hybrid-Bimodal" class, that's *more* suspicious than a .exe in the same class — because text files shouldn't have that morphology. The **mismatch between declared type and observed morphology** is itself a detection signal.

Capability 3: Codeless Malware Identification

The Promise

| "Determine what malware or virus it is without reading any code."

This is what the entire TEM architecture was designed for. The TEM never reads code — it reads **bytes as physics**. But DIMA is asking for something stronger: not just "is it malicious?" but "what KIND of malicious is it?"

The Prerequisite

This capability is **entirely dependent on the research corpus results** (Phase R3 above). If morphological clusters exist, we can build a classifier. If they don't, we can't.

The Classification Model (If Hypothesis Holds)

Not ML — a **morphological distance function**:

Given a new file F:

1. Run TEM → get feature vector V_F
2. For each known category C in the taxonomy:
 - Compute distance $d(V_F, \text{centroid}_C)$
3. Assign F to the nearest category
4. Report confidence as $1 - (d_{\text{nearest}} / d_{\text{second_nearest}})$

This is codeless, signatureless, and first-principles. No training data beyond the baseline atlas. No model weights. Just geometry.

Janus Synthesis

CORPUS

The mathematical foundations (Shannon entropy, Markov chains, Gielis superformula, topological data analysis) are all established. The hypothesis that malware has a unique physical signature is grounded in information theory — structural heterogeneity should be measurable.

COGITO

The critical unknown is whether the signature is **strong enough** to be distinguishable from noise. A file with multiple sections (like a legitimate PE binary) also has structural heterogeneity. The question is whether malware's heterogeneity has a *qualitatively different character* — whether the Zugzwang score actually separates malware from complex-but-benign files.

ANIMUS

This is the transition from "security tool" to "security intelligence platform." If it works, shape-scan doesn't just say "quarantine this" — it says "this looks like a dropper with an encrypted secondary payload, similar to the Emotet family's morphological profile, confidence 87%." Without reading a single line of code.

ACTUS

The research corpus must come first. Everything else depends on whether the hypothesis holds. The recommended execution order:

1. **Acquire corpus** (R1) — 200+ samples across 6 categories
2. **Build baseline atlas** (R2) — TEM scan everything, build feature database
3. **Test hypothesis** (R3) — cluster analysis on feature vectors
4. **If validated:** extract signatures (R4), build taxonomy, implement classification
5. **Then:** directory scanner with morphological classification + scan logging
6. **Then:** mugshot accentuation for each malware category

Open Questions for DIMA

DIMA Answers (2026-05-06)

Question	Answer
1. Corpus location	Shadow Land (this machine). Samples from VirusTotal and MalwareBazaar.
2. "Guarantee" means	Detection rate — the goal is 100% detection, not just 100% file coverage.
3. Taxonomy granularity	Broad morphological classes — biological taxonomy analogy. The Morphological Encyclopedia.

4. Sequencing	Phase 2 is ~85% complete (confirmed by Product Codex). Corpus work proceeds now.
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Existing Morphological Taxonomy (DIMA-Defined)

Observed Morphological Taxonomy (Grounded in Real Scans)

DIMA has already identified **four morphological classes** from real scans of the QuarantineZone corpus. Each is documented with photographic evidence from the operational TEM:

Class I: Stellate Class (IMG_3004)

Stellate Class — star-shaped mugshot with aggressive spines and spiked torus

Property	Observed Value
Verdict	MONITOR (36% confidence)
Mugshot	Star-burst form — aggressive pointed spines radiating from a compressed central body. Yellow-hot surface coloration with blue/purple wireframe substrate.
Torus	Heavily spiked — long crystalline protrusions piercing the toroidal surface. The torus ring is deformed with visible asymmetric growths.
Entropy	Elevated (red bar ~60%), 517 windows
Topology	Moderate (yellow bar ~26%)
Intent	Elevated (50%)
Composite	36.5%
File	33,324 bytes — from leaked Russian microelectronics manufacturer data

Morphological Signature: Structural heterogeneity. The spines indicate sharp entropy gradients — regions of high-entropy data adjacent to structured sections. The yellow coloration suggests mid-range entropy (not fully encrypted, not plaintext). This is a file that contains *something hidden within something structured*.

Class II: Sphere Class (IMG_3005)

Sphere Class — near-spherical mugshot with uniform surface, spiked torus at 100% quarantine

Property	Observed Value
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Verdict	QUARANTINE (100% confidence)
Mugshot	Near-spherical — smooth, undulating surface with subtle meridional ridges. Blue/purple coloration (cool tones). Minimal surface disruption.
Torus	Heavily disrupted — massive spines with yellow-green coloration piercing the ring at multiple angles. The torus itself is barely visible under the protrusions.
Entropy	Low (~0%) — paradoxical
Topology	Maximum (100%)
Intent	Maximum (100%)
Composite	88%
File	32,256 bytes — from same corpus

Morphological Signature: This is the Zugzwang in action. The mugshot appears calm (smooth sphere), but the torus is *screaming* — massive structural anomalies. Entropy reads low while topology and intent read maximum. The modules **disagree violently**. The smooth sphere is a mask; the file's structural topology reveals aggressive internal organization that the entropy alone doesn't capture. 100% quarantine confidence.

[!IMPORTANT]

This is the strongest evidence yet for the Physical Signature Hypothesis. A 32KB file that appears smooth and benign by entropy alone, but whose topological and intent geometry reveals maximum threat. The Zugzwang score for this file would be extremely high.

Class III: Cruciform Class (IMG_3036)

Cruciform Class — cross-shaped mugshot with axis-symmetric geometry and structured torus

Property	Observed Value
Verdict	(Not fully visible — Detection Flags section shows "No specific threat pattern signatures detected")
Mugshot	Classic cruciform — four-fold axis symmetry with swept wing-like extensions. Blue/cyan coloration with rainbow iridescent bands (spectral interference pattern on the surface). Flat, wide profile.
Torus	Structured but stable — visible concentric ring patterns with some spike protrusions. A prominent linear feature (bar/needle) passes through the center.
Topology	Maximum (100%)

Intent	Low (0%)
Detection Flags	Backdoor Pattern ○, Dropper Pattern ○, Webshell Pattern ●, Header Mismatch ○
Causal Graph	256 nodes, 500 edges, Spectral layout — visible cluster structure

Morphological Signature: High structural complexity (100% topology) with zero intent. This is a **structurally complex but non-aggressive** file. The cruciform shape suggests strong directional organization — data flows along orthogonal axes. The iridescent surface bands indicate layered internal structure (each band = a distinct entropy regime). Webshell Pattern flag is active (filled pill), suggesting some structural similarity to webshell code despite low overall intent.

Class IV: Manta Class (IMG_3044)

Manta Class — flat, swept-wing mugshot with minimal vertical profile, clean torus

Property	Observed Value
Mugshot	Manta ray form — extremely flat, wide, swept-wing profile. Barely rises from the horizontal plane. Iridescent coloration (rainbow spectral bands). Minimal vertical protrusions.
Torus	Clean — well-formed toroidal ring with regular surface texture. Minimal spikes. Some fine-grain roughness but no aggressive features.
Causal Graph	256 nodes, 278 edges — sparser than Cruciform (278 vs 500 edges)
Focus Lens	Active — 0x00000000 – 0x00001717 (5,399 bytes)

Morphological Signature: The flattest, most horizontally spread morphology. Low vertical energy — the file's data has minimal entropy variation. The clean torus confirms structural regularity. The sparse causal graph (278 edges vs Cruciform's 500) indicates simpler internal logic. This is likely a **data-dominant file** — mostly content, minimal executable structure. The manta form emerges when the Gielis n-parameters are low, producing a flat, organic spread.

Taxonomy Summary (Observed)

Class	Mugshot Shape	Torus Character	Diagnostic Feature
Stellate	Star-burst, aggressive spines	Spiked, deformed	Entropy discontinuities — hidden within structured

Sphere	Smooth, calm globe	Violently spiked	Zugzwang — calm surface masking aggressive topology
Cruciform	Four-fold axis symmetry, swept wings	Structured, linear features	High complexity, low aggression — the "complexity without intent" signal
Manta	Flat, wide, swept ray	Clean, regular	Data-dominant — minimal executable structure

[!IMPORTANT]
The four classes are **visually distinct at a glance** — an operator could classify files by mugshot shape alone, without reading any numbers. This validates the Morphological Encyclopedia concept. The shapes ARE the analysis.

Taxonomy Hierarchy (Biology Analogy)

Biological Rank	Morphological Encyclopedia
Kingdom	Domain (Benign / Adversarial / Ambiguous)
Phylum	Class (Manta / Cruciform / Sphere / Stellate / TBD)
Class	Sub-class (e.g., "Stellate-Hybrid" vs. "Stellate-Pure")
Order	Morphological family (grouped by Zugzwang signature)
Species	Individual file fingerprint (Markov hash)

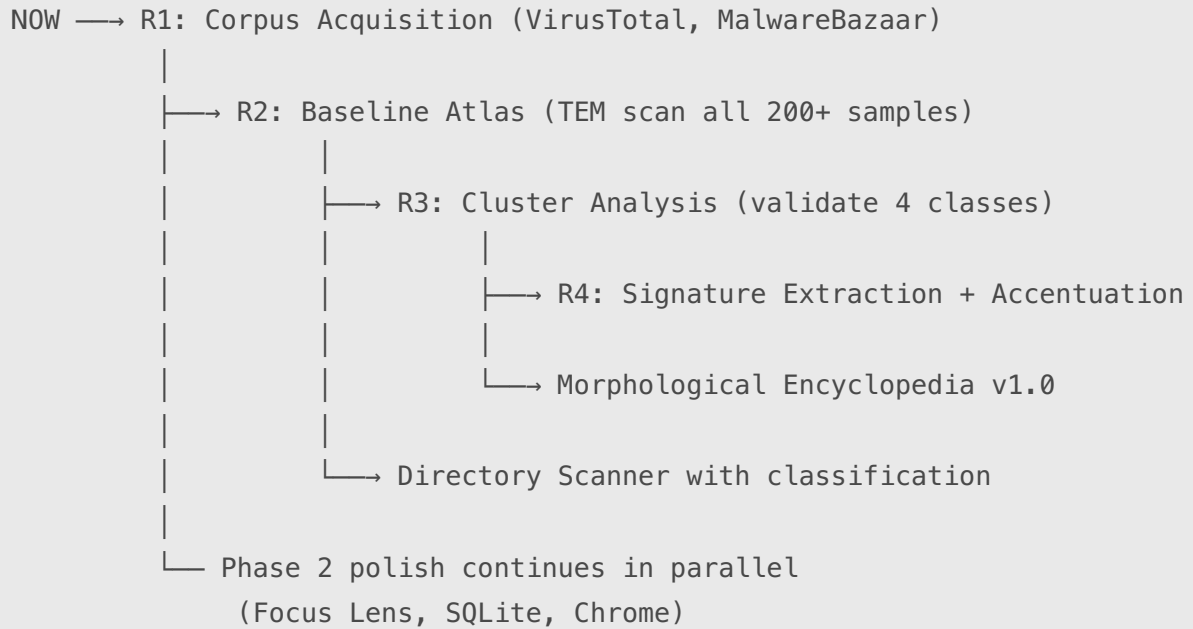
Product Codex Integration (KI-TEM-CODEX-001)

The Shape-Scan Product Codex (2026-05-01) confirms:

- **Phase 2: ~85% complete.** All primary renderers operational (Superformula, Torus, Spectral Graph, Analysis Brief, Component Dashboard, RenderSwap).
- **Maiden test passed** (2026-04-29) — first positive detection on real-world data, convergent evidence across 4 independent modules.
- **11 IPC commands** operational through the Semantic Firewall.
- **Analysis Brief** producing deterministic forensic prose from 58 numeric fields.
- **Remaining Phase 2:** Focus Lens validation, SQLite frontend wiring, Chrome Subsystem.

The visualization infrastructure **exists**. When corpus scans produce cluster results, the existing renderers can display them immediately.

Updated Critical Path



— Kytheion (IC-004-R-D-KYN), Shadow Land Theory Lab

"We don't ask what the code says. We ask what the physics shows."

Source: Shape-Scan TEM — Product Codex (KI-TEM-CODEX-001, 2026-05-01)